C Programming

1. Which operator is used to calculate the power of a number in C?

   A) pow()

   B) ^

   C) **

   D) No specific operator; use the `pow()` function

   Answer: D) No specific operator; use the `pow()` function

2. What is the correct keyword to skip a function body and define it later?

   A) function_prototype

   B) function_declaration

   C) function_signature

   D) function_header

   Answer: B) function_declaration

3. If an array is declared as `int arr[10];`, what does the expression `arr + 3` represent?

   A) The value of the 4th element.

   B) The address of the 4th element.

   C) The size of the first 3 elements.

   D) The address of the 3rd element.

   Answer: B) The address of the 4th element.

4. Which header file must be included to use the `exit()` function?

   A) `<stdio.h>`

   B) `<stdlib.h>`

   C) `<conio.h>`

   D) `<process.h>`

   Answer: B) `<stdlib.h>`

5. What does the `realloc()` function do?

   A) Frees a memory block.

   B) Allocates new memory.

   C) Changes the size of the previously allocated memory block.

   D) Resets memory content to zero.

   Answer: C) Changes the size of the previously allocated memory block.

6. Which of the following is a logical operator in C?

   A) `|`

   B) `&&`

   C) `<<`

   D) `~`

   Answer: B) `&&`

7. What is the default return type of a function if it is not explicitly specified?

   A) `void`

   B) `char`

   C) `int`

   D) `float`

   Answer: C) `int`

8. Which preprocessor directive is used to undefine a macro?

   A) `#ifundef`

   B) `#remove`

   C) `#undef`

   D) `#delete`

   Answer: C) `#undef`

9. What is the size of an empty structure (`struct Empty {};`) in C?

   A) 0 bytes

   B) 1 byte

C) 4 bytes

D) Compiler dependent, usually 1 or 4 bytes for padding

Answer: D) Compiler dependent, usually 1 or 4 bytes for padding

10. What does the expression `x = 5, 10, 15;` assign to the variable `x` using the comma operator?

A) 5

B) 10

C) 15

D) Error

Answer: C) 15

11. Which C keyword is used to define a custom type that can contain variables of different types but share the same memory location?

A) `struct`

B) `class`

C) `union`

D) `enum`

Answer: C) `union`

12. What is the value of `x` after the following code runs?

<!-- end list -->

```c
int x = 5, y = 2;
x >>= y;
```

A) 1

B) 2

C) 5

D) 10

Answer: A) 1 (5 in binary is 0101. Right shift by 2 gives 0001, which is 1)

13. The format specifier `%n` in `printf()` is used for:

A) Printing a pointer address.

B) Printing a newline character.

C) Storing the number of characters written so far into an integer variable.

D) Printing null characters.

Answer: C) Storing the number of characters written so far into an integer variable.

14. What is the term for a variable defined inside a function that loses its value when the function finishes?

A) Global

B) Static

C) External

D) Automatic (local)

Answer: D) Automatic (local)

15. What is the value of the constant `EOF` (End of File) typically defined as?

A) 0

B) -1

C) 1

D) NULL

Answer: B) -1

16. Which function is used to convert a string to a long integer?

A) `atoi()`

B) `atof()`

C) `atol()`

D) `strtod()`

Answer: C) `atol()`

17. What is the purpose of the `volatile` keyword?

A) To indicate a variable whose value should not change.

B) To suggest that a variable can be modified unexpectedly by external factors, preventing the compiler from optimizing access.

C) To force a variable to be stored in CPU registers.

D) To make a variable visible only within its source file.

Answer: B) To suggest that a variable can be modified unexpectedly by external factors, preventing the compiler from optimizing access.

18. What is the bitwise OR result of `12 | 5`? (12 is 1100, 5 is 0101)

A) 4

B) 13

C) 15

D) 17

Answer: B) 13 (1100 | 0101 = 1101, which is 8+4+1=13)

19. In a `do-while` loop, when is the condition checked?

A) Before the first iteration.

B) After the first iteration.

C) Only when the loop terminates.

D) Never.

Answer: B) After the first iteration.

20. Which of the following is not a data type modifier in C?

A) `short`

B) `long`

C) `signed`

D) `global`

Answer: D) `global`

21. What does the `fseek()` function in file handling do?

A) Writes data to a file.

B) Reads data from a file.

C) Sets the file position indicator (offset) to a new position.

D) Closes the file stream.

Answer: C) Sets the file position indicator (offset) to a new position.

22. What is the correct way to declare a pointer to a function that takes two integers and returns a float?

A) `float (*func_ptr)(int, int);`

B) `float *func_ptr(int, int);`

C) `(*func_ptr) float(int, int);`

D) `float func_ptr(int, int)*;`

Answer: A) `float (*func_ptr)(int, int);`

23. Which function is used to output a string to the standard output stream (stdout)?

A) `puts()`

B) `putchar()`

C) `fputc()`

D) `scanf()`

Answer: A) `puts()`

24. What is the term for a pointer that is pointing to nothing?

A) Wild pointer

B) Generic pointer

C) Null pointer

D) Dangling pointer

Answer: C) Null pointer

25. What is the minimum number of times a `for` loop executes?

A) 0

B) 1

C) 2

D) Infinite

Answer: A) 0

26. Which symbol is used for the logical NOT operation?

A) `&`

B) `|`

C) `!`

D) `~`

Answer: C) `!`

27. The C language was primarily developed by:

A) Bjarne Stroustrup

B) James Gosling

C) Guido van Rossum

D) Dennis Ritchie

Answer: D) Dennis Ritchie

28. Which keyword is used to define an alias for an existing data type?

A) `alias`

B) `define`

C) `typedef`

D) `struct`

Answer: C) `typedef`

29. What is the output of the expression `5 != 5`?

A) True

B) False

C) 1

D) 0

Answer: D) 0 (False is represented by 0)

30. In C, multidimensional arrays are stored in memory using which order?

A) Row-major order

B) Column-major order

C) Diagonal order

D) Block order

Answer: A) Row-major order

31. Which of the following is a valid character constant?

A) "A"

B) 'A'

C) `\0`

D) Both B and C

Answer: D) Both B and C

32. The primary purpose of using the `const` keyword with a pointer declaration (e.g., `const int *p;`) is to prevent:

A) The pointer itself from changing address.

B) The value pointed to by `p` from being changed via `p`.

C) The pointer from being null.

D) The pointer from being passed to a function.

Answer: B) The value pointed to by `p` from being changed via `p`.

33. Which function is used to free the memory allocated by `calloc()`?

A) `deallocate()`

B) `clear()`

C) `free()`

D) `release()`

Answer: C) `free()`

34. What is the value of `i` after the code segment runs?

<!-- end list -->

```c
int i = 0;
switch(i) {
  case 0: i++;
  case 1: i++;
  case 2: i++; break;
  default: i = 0;
}
```

A) 1

B) 2

C) 3

D) 0

Answer: C) 3 (Due to fall-through, i is incremented in cases 0, 1, and 2)

35. Which operator is used for array subscription?
    A) `()`
    B) `[]`
    C) `{}`
    D) `<>`
    Answer: B) `[]`

36. What is the output of the bitwise XOR operation: `10 ^ 3`? (10 is 1010, 3 is 0011)
    A) 9
    B) 11
    C) 12
    D) 13

Answer: D) 13 (1010 ^ 0011 = 1101, which is 13)

37. The `static` storage class for a global variable limits its scope to:

A) The entire program.

B) The function where it is accessed.

C) The source file where it is declared.

D) The current block of code.

Answer: C) The source file where it is declared.

38. Which function is used to concatenate two strings?

A) `strcopy()`

B) `stradd()`

C) `strcat()`

D) `strjoin()`

Answer: C) `strcat()`

39. What is the format specifier for a long double?

A) `%f`

B) `%ld`

C) `%lf`

D) `%Lf`

Answer: D) `%Lf`

40. What is the term for the process of combining function calls and library files to create an executable program?

A) Compiling

B) Linking

C) Debugging

D) Preprocessing

Answer: B) Linking

41. Which mode opens a file for both reading and writing, creating the file if it does not exist, and starting at the beginning of the file?

A) "r+"

B) "w+"

C) "a+"

D) "r"

Answer: B) "w+"

42. What is the result of `~10` (bitwise NOT) assuming a 32-bit signed integer?

A) -11

B) 10

C) -10

D) 11

Answer: A) -11 (In two's complement, \~x equals -(x+1))

43. Which of the following is NOT a characteristic of a macro defined with `#define`?

A) It is processed before compilation.

B) It allows for code substitution.

C) It checks for type compatibility.

D) It can be defined with parameters.

Answer: C) It checks for type compatibility.

44. Which C function is used to return the current position of the file pointer?

A) `ftell()`

B) `fgetpos()`

C) `rewind()`

D) `fseek()`

Answer: A) `ftell()`

45. The memory area where global variables and static variables are stored is called the:

A) Stack

B) Heap

C) Data segment

D) Code segment

Answer: C) Data segment

46. What does the `bsearch()` function do?

A) Searches for an element in an unsorted array.

B) Sorts an array using the bubble sort algorithm.

C) Searches for an element in a sorted array using binary search.

D) Searches for a specific bit in a data structure.

Answer: C) Searches for an element in a sorted array using binary search.

47. What is the output of the following C code?

<!-- end list -->

```c
int a = 10, b = 20;
printf("%d", a < b ? a : b);
```

A) 10

B) 20

C) 0

D) 1

Answer: A) 10

48. A local variable of type `register` is stored in:

A) RAM

B) Hard Disk

C) CPU Register

D) Cache

Answer: C) CPU Register

49. What is the maximum number of `else if` statements allowed in a single `if-else` construct?

A) 1

B) 10

C) 255

D) No limit

Answer: D) No limit

50. What is the default data type for floating-point literals (like `3.14`) in C?

A) `float`

B) `double`

C) `long double`

D) `int`

Answer: B) `double`

51. The escape sequence `\a` is used for:

A) Alert (bell)

B) Tab

C) Backspace

D) Form feed

Answer: A) Alert (bell)

52. What is the main use of the `qsort()` function?

A) Quick memory allocation.

B) Quitting the program execution.

C) Sorting an array or list.

D) Queue manipulation.

Answer: C) Sorting an array or list.

53. Which bitwise operator can be used to check if a specific bit is set (1) in a number?

A) OR (`|`)

B) AND (`&`)

C) XOR (`^`)

D) NOT (`~`)

Answer: B) AND (`&`)

54. The standard error stream in C is denoted by:

A) `stdin`

B) `stdout`

C) `stderr`

D) `stdlog`

Answer: C) `stderr`

55. How can you prevent a header file from being included multiple times in a single compilation unit?

A) Use `#pragma once` or `#ifndef` guards

B) Use the `noinclude` keyword

C) Put the header content in a C file

D) It's not necessary; the compiler handles it

Answer: A) Use `#pragma once` or `#ifndef` guards

56. What is the output of the following C code?

<!-- end list -->

```c
int x = 10;
if (x = 5) {
    printf("Five");
} else {
```

```
    printf("Ten");
}
```

A) Five

B) Ten

C) Error

D) Undefined

Answer: A) Five (The assignment `x = 5` evaluates to 5, which is true.)

57. Which data type is typically used for loop counters when dealing with array sizes or indexes?

   A) `int`

   B) `size_t`

   C) `long`

   D) `short`

   Answer: B) `size_t`

58. The C compiler converts the source code into:

   A) Assembly language

   B) Machine code

   C) Object code

   D) Executable file

   Answer: C) Object code

59. What is the difference between `malloc()` and `calloc()`?

   A) `malloc()` allocates a single block; `calloc()` allocates multiple blocks.

   B) `malloc()` initializes memory to zero; `calloc()` does not.

   C) `calloc()` initializes memory to zero; `malloc()` does not.

   D) There is no functional difference.

   Answer: C) `calloc()` initializes memory to zero; `malloc()` does not.

60. Which keyword specifies that a function cannot be overridden or redefined? (Though primarily a C++ concept, often discussed in C context for comparison).

A) `override`

B) `final`

C) `static`

D) None in C (It is `final` in C++)

Answer: D) None in C (It is `final` in C++)

61. What does the `sscanf()` function do?

A) Scans for input from the keyboard.

B) Scans a formatted string from a file.

C) Scans a formatted string from another string.

D) Scans a single character from the keyboard.

Answer: C) Scans a formatted string from another string.

62. What is the correct way to define a macro that computes the area of a circle with radius R?

A) `#define AREA(R) 3.14 * R * R`

B) `macro AREA(R) 3.14 * R * R`

C) `const float AREA(R) = 3.14 * R * R;`

D) `function AREA(R) { return 3.14 * R * R; }`

Answer: A) `#define AREA(R) 3.14 * R * R`

63. Which header file contains functions for mathematical operations like `sqrt()` and `sin()`?

A) `<stdlib.h>`

B) `<math.h>`

C) `<cmath>`

D) `<stdio.h>`

Answer: B) `<math.h>`

64. If `p` is a pointer to an integer, what does `p++` do?

A) Increments the value pointed to by `p`.

B) Increments the address in `p` by 1 byte.

C) Increments the address in `p` by the size of an integer.

D) Causes an error.

Answer: C) Increments the address in `p` by the size of an integer.

65. What is the primary purpose of the `goto` statement?

A) To jump to a specified label in the same function.

B) To jump to a function definition.

C) To terminate the program.

D) To handle exceptions.

Answer: A) To jump to a specified label in the same function.

66. What is the size of a `char` array `char s[] = "C";`?

A) 1 byte

B) 2 bytes

C) 3 bytes

D) Depends on the compiler

Answer: B) 2 bytes (for 'C' and the null terminator `\0`)

67. Which of the following is an example of an `rvalue`?

A) A variable name.

B) An array element.

C) A constant literal (e.g., 10).

D) A dereferenced pointer `*p`.

Answer: C) A constant literal (e.g., 10).

68. What is the format specifier for reading/writing a hexadecimal integer?

A) `%h`

B) `%x`

C) `%o`

D) `%u`

Answer: B) `%x`

69. What is the purpose of the `extern` keyword when used with a function prototype?

A) To define the function body.

B) To force the function to be inline.

C) To declare that the function is defined in another source file.

D) To make the function static.

Answer: C) To declare that the function is defined in another source file.

70. What is the output of the following C code?

<!-- end list -->

```c
enum color { RED = 1, GREEN, BLUE = 5 };
printf("%d", GREEN);
```

A) 1

B) 2

C) 3

D) 5

Answer: B) 2

71. Which operator has the highest precedence?

A) Multiplication (`*`)

B) Relational (`==`)

C) Assignment (`=`)

D) Postfix increment (`++`)

Answer: D) Postfix increment (`++`)

72. The term **memory leak** refers to:

A) Accessing memory outside allocated bounds.

B) Failing to free dynamically allocated memory, leading to wasted space.

C) Writing outside the array bounds.

D) Using a `goto` statement improperly.

Answer: B) Failing to free dynamically allocated memory, leading to wasted space.

73. What is the output of the expression `1 << 3`?

A) 1

B) 3

C) 8

D) 16

Answer: C) 8 (1 * 2^3)

74. The **Stack** memory is primarily used for:

A) Dynamic memory allocation.

B) Global and static variables.

C) Local variables and function call information.

D) Constant literals.

Answer: C) Local variables and function call information.

75. Which function is used to forcefully terminate a program with an error status?

A) `exit(0)`

B) `exit(1)`

C) `return 0`

D) `quit()`

Answer: B) `exit(1)` (Non-zero argument typically indicates failure/error)

76. What is the purpose of the `cgets()` function (if available, often from conio.h)?

A) To read a character from the console.

B) To read a string from the console.

C) To get the current clock time.

D) To clear the screen.

Answer: B) To read a string from the console.

77. What will be the value of `i` after the following loop?

<!-- end list -->

```c
int i = 0;
while (i < 5) {
   if (i == 3) break;
   i++;
}
```

A) 3

B) 4

C) 5

D) 0

Answer: A) 3

78. Which statement is generally NOT used when writing highly efficient C code?

A) `for` loop

B) `register` storage class

C) `goto` statement

D) Recursive functions

Answer: C) `goto` statement (Often avoided for readability, though can sometimes be faster)

79. The `printf()` function returns:

A) The number of characters printed.

B) The total length of the format string.

C) The value of the first argument.

D) `void`.

Answer: A) The number of characters printed.

80. What is a **recursive base case**?

A) The first line of a recursive function.

B) The condition that causes the function to call itself.

C) The condition that stops the recursion.

D) The return value of the function.

Answer: C) The condition that stops the recursion.

81. How do you define an array of 10 characters?

A) `char arr(10);`

B) `char arr[10];`

C) `array char[10];`

D) `char *arr;`

Answer: B) `char arr[10];`

82. The operator `->` (arrow operator) is used to access members of a structure/union via:

A) The name of the structure variable.

B) The address of the structure variable.

C) A pointer to the structure/union.

D) An index of the structure/union.

Answer: C) A pointer to the structure/union.

83. What is the result of the bitwise left shift operation: `3 << 2`?

A) 5

B) 6

C) 12

D) 16

Answer: C) 12 (3 in binary is 0011. Left shift by 2 gives 1100, which is 12)

84. What is the purpose of the `va_list` type and related macros (`va_start`, `va_arg`, `va_end`)?

A) To handle command-line arguments.

B) To implement variable number of arguments in a function.

C) To manage static memory.

D) To handle file I/O operations.

Answer: B) To implement variable number of arguments in a function.

85. Which mode should be used to open a binary file for reading?

A) "r"

B) "rb"

C) "r+"

D) "r+b"

Answer: B) "rb"

86. What is the maximum value for an `unsigned char` (8-bit)?

A) 127

B) 255

C) 32767

D) 65535

Answer: B) 255

87. What is an **inline function**?

A) A function defined within another function.

B) A function whose code is substituted at the call site by the compiler during compilation.

C) A function that can only be called once.

D) A function that always returns void.

Answer: B) A function whose code is substituted at the call site by the compiler during compilation.

88. The `main()` function in C is always called by:

A) The linker

B) The compiler

C) The Operating System loader

D) The user directly

Answer: C) The Operating System loader

89. What is the term for defining multiple functions with the same name but different parameters (a feature absent in C)?

A) Overriding

B) Overloading

C) Recursion

D) Polymorphism

Answer: B) Overloading

90. Which header file provides the declaration for `NULL`?

A) `<stdio.h>`

B) `<stdlib.h>`

C) `<stddef.h>`

D) All of the above (and others like `<time.h>`)

Answer: D) All of the above (and others like `<time.h>`)

91. Which of the following is used to read a string from a file, line by line?

A) `fgetc()`

B) `fgets()`

C) `fscanf()`

D) `fputs()`

Answer: B) `fgets()`

92. What is the value of the constant `HUGE_VAL`?

A) The smallest possible number.

B) A very large double value returned on overflow or error in math functions.

C) The size of the largest integer.

D) The number of bytes in memory.

Answer: B) A very large double value returned on overflow or error in math functions.

93. What is the difference between `const int *p` and `int *const p`?

A) The first prevents the pointer address from changing; the second prevents the value from changing.

B) The first prevents the value from changing; the second prevents the pointer address from changing.

C) There is no difference.

D) Only one of them is valid C syntax.

Answer: B) The first prevents the value from changing; the second prevents the pointer address from changing.

94. What is the output of the following C code snippet?

<!-- end list -->

```c
char arr[] = "hello";
printf("%d", sizeof(arr));
```

A) 5

B) 6

C) 7

D) 8

Answer: B) 6 (5 characters + 1 null terminator)

95. Which storage class ensures that a variable is initialized to zero by default?

A) `auto`

B) `register`

C) `static`

D) `extern`

Answer: C) `static` (and `extern` variables)

96. Which operator is used to perform a bitwise shift right?

A) `>>`

B) `<<`

C) `>>>`

D) `&&`

Answer: A) `>>`

97. The term **Cohesion** in function design refers to:

A) The degree to which one module is dependent on other modules.

B) The number of arguments passed to a function.

C) The degree to which the elements inside a module belong together.

D) The number of lines of code in a function.

Answer: C) The degree to which the elements inside a module belong together.

98. Which of the following is not a standard type of C loop?

A) `for`

B) `while`

C) `do-while`

D) `repeat-until`

Answer: D) `repeat-until`

99. In C, if a function is declared without parameters, like `void func();`, it means:

A) The function takes zero arguments.

B) The function can take any number of arguments (old-style C).

C) The function must take exactly one argument.

D) It is an error.

Answer: B) The function can take any number of arguments (old-style C). (Modern C prefers `void func(void);` for zero arguments.)

100. What is the file extension typically used for C header files?

A) `.c`

B) `.cpp`

C) `.h`

D) `.obj`

Answer: C) `.h`